

## Project Planning Phase

### Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

|               |                                                                                              |
|---------------|----------------------------------------------------------------------------------------------|
| Date          | 21 June 2025                                                                                 |
| Team ID       | LTVIP2025TMID41595                                                                           |
| Project Name  | GrainPalette - A Deep Learning Odyssey In Rice Type Classification Through Transfer Learning |
| Maximum Marks | 5 Marks                                                                                      |

### Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create product backlog and sprint schedule

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                          | Story Points | Priority | Team Members                                                                                        |
|----------|-------------------------------|-------------------|--------------------------------------------------------------------------------------------|--------------|----------|-----------------------------------------------------------------------------------------------------|
| Sprint-1 | Image data collection         | USN-1             | As a user, I want to collect and label different rice grain images for training the model. | 3            | High     | 1. N Karthik<br><br>2. N Jagadeeswar Reddy<br><br>3. N Venkateswara Reddy<br><br>4.Ns Iffath Muskan |

|          |                    |       |                                                                                                         |   |      |                                                                           |
|----------|--------------------|-------|---------------------------------------------------------------------------------------------------------|---|------|---------------------------------------------------------------------------|
| Sprint-1 | Data preprocessing | USN-2 | As a developer, I want to preprocess the rice images (resize, normalization) to improve model accuracy. | 2 | High | 1. N Karthik<br><br>2. N Jagadeeswar Reddy<br><br>3. N Venkateswara Reddy |
|----------|--------------------|-------|---------------------------------------------------------------------------------------------------------|---|------|---------------------------------------------------------------------------|

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                                                                         | Story Points | Priority | Team Members                                                                                        |
|----------|-------------------------------|-------------------|-----------------------------------------------------------------------------------------------------------|--------------|----------|-----------------------------------------------------------------------------------------------------|
|          |                               |                   |                                                                                                           |              |          | 4.Ns Iffath Muskan                                                                                  |
| Sprint-2 | Model traning                 | USN-3             | As a data scientist, I want to apply transfer learning on a pre-trained CNN model to classify rice types. | 5            | high     | 1. N Karthik<br><br>2. N Jagadeeswar Reddy<br><br>3. N Venkateswara Reddy<br><br>4.Ns Iffath Muskan |

|               |                                      |                          |                                                                                                                   |                     |                 |                                                                                         |
|---------------|--------------------------------------|--------------------------|-------------------------------------------------------------------------------------------------------------------|---------------------|-----------------|-----------------------------------------------------------------------------------------|
| Sprint-2      | Model evaluation                     | USN-4                    | As a developer, I want to evaluate the model using accuracy, confusion matrix, and classification report.         | 2                   | medium          | 1. N Karthik<br>2. N Jagadeeswar Reddy<br>3. N Venkateswara Reddy<br>4.Ns Iffath Muskan |
| Sprint-3      | Ui development                       | USN-5                    | As a user, I want to upload a rice grain image through a simple web or mobile app and get classification results. | 3                   | medium          | 1. N Karthik<br>2. N Jagadeeswar Reddy<br>3. N Venkateswara Reddy<br>4.Ns Iffath Muskan |
| <b>Sprint</b> | <b>Functional Requirement (Epic)</b> | <b>User Story Number</b> | <b>User Story / Task</b>                                                                                          | <b>Story Points</b> | <b>Priority</b> | <b>Team Members</b>                                                                     |
|               |                                      |                          |                                                                                                                   |                     |                 | 1. N Karthik<br>2. N Jagadeeswar Reddy                                                  |

|           |            |       |                                                                                                         |   |     |                                                                                         |
|-----------|------------|-------|---------------------------------------------------------------------------------------------------------|---|-----|-----------------------------------------------------------------------------------------|
|           |            |       |                                                                                                         |   |     | 3. N Venkateswara Reddy<br>4.Ns Iffath Muskan                                           |
| Sprint -3 | Deployment | USN-6 | As a developer, I want to deploy the trained model and integrate it with the frontend for realtime use. | 4 | low | 1. N Karthik<br>2. N Jagadeeswar Reddy<br>3. N Venkateswara Reddy<br>4.Ns Iffath Muskan |

**Project Tracker, Velocity & Burndown Chart: (4 Marks)**

| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) | Sprint Release Date (Actual) |
|----------|--------------------|----------|-------------------|---------------------------|-------------------------------------------------|------------------------------|
| Sprint-1 | 20                 | 6 Days   | 15 June 2025      | 20 June 2025              | 20                                              | 20 June 2025                 |

|               |                           |                 |                          |                                  |                                                        |                                     |
|---------------|---------------------------|-----------------|--------------------------|----------------------------------|--------------------------------------------------------|-------------------------------------|
| Sprint-2      | 20                        | 6 Days          | 17 June 2025             | 22 June 2025                     | 20                                                     | 22 June 2025                        |
| Sprint-3      | 20                        | 6 Days          | 19 June 2025             | 24 June 2025                     | 20                                                     | 24 June 2025                        |
| <b>Sprint</b> | <b>Total Story Points</b> | <b>Duration</b> | <b>Sprint Start Date</b> | <b>Sprint End Date (Planned)</b> | <b>Story Points Completed (as on Planned End Date)</b> | <b>Sprint Release Date (Actual)</b> |
| Sprint-4      | 20                        | 6 Days          | 21 June 2025             | 26 June 2025                     | 20                                                     | 26 June 2025                        |
|               |                           |                 |                          |                                  |                                                        |                                     |
|               |                           |                 |                          |                                  |                                                        |                                     |
|               |                           |                 |                          |                                  |                                                        |                                     |
|               |                           |                 |                          |                                  |                                                        |                                     |

#### Velocity:

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

**Burndown Chart:**

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

**Reference:**

<https://www.atlassian.com/agile/project-management>

<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>

<https://www.atlassian.com/agile/tutorials/epics>

<https://www.atlassian.com/agile/tutorials/sprints>

<https://www.atlassian.com/agile/project-management/estimation>

<https://www.atlassian.com/agile/tutorials/burndown-charts>